

# HARSHIT TIWARI

Postdoctoral Associate, New York University

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## RESEARCH INTERESTS

Turbulent convection, atmospheric and astrophysical flows, compressible flows, high-performance computing (HPC), machine learning, quantum computing, quantum turbulence, and nonlinear dynamics.

## EDUCATION

-  **Indian Institute of Technology Kanpur** Kanpur, India  
*Doctor of Philosophy in Physics* **August 2021 - February 2026**
  - Courses on Field Theory, Physics of Turbulence, High-Performance Computing and Advanced Statistical Physics.
  - Cumulative Performance Index: 9.56/10
-  **Indian Institute of Technology Kanpur** Kanpur, India  
*Master of Science in Physics* **July 2019 - July 2021**
  - Courses on High Energy Astrophysics, Nuclear and Particle Physics and Quantum Field Theory.
  - Cumulative Performance Index: 8.30/10
-  **Kumaun University** Nainital, India  
*Bachelor of Science* **July 2016 - June 2019**
  - Subjects: Physics, Mathematics, Chemistry
  - Percentage: 66.4%, First Class

## RESEARCH EXPERIENCE

-  **Postdoctoral Associate** **July 2026 - Present**  
**Department of Mechanical and Aerospace Engineering**  
**Tandon School of Engineering, New York University**  
Supervisor: Prof. Katepalli R. Sreenivasan, New York University  
Project: **Quantum Computing for Fluid Dynamics (QCFD)**
-  **Fellowship for Academic and Research Excellence (FARE)** **March 2026 - June 2026**  
**Department of Physics, Indian Institute of Technology Kanpur**
  - High-resolution simulations of moist turbulent convection in Earth's troposphere.
  - Thermalisation in quantum and classical Euler turbulence.
  - Transients in thermal convection and energy distribution among various scales.
-  **Graduate Researcher** **August 2021 - February 2026**  
**Department of Physics, Indian Institute of Technology Kanpur**  
Supervisor: Prof. Mahendra Verma, Department of Physics, IIT Kanpur  
Co-supervisor: Prof. Rajesh Ranjan, Department of Aerospace Engineering, IIT Kanpur  
Thesis Title: **Compressible Turbulent Convection at Extreme Rayleigh Numbers**
  - Simulated turbulent compressible convection at extreme Rayleigh numbers, revealing classical heat transport scaling laws.
  - Developed a scalable Python PDE solver with GPU and MPI support for high-performance simulations.
  - Applied advanced numerical methods to study shocks, turbulence, and compressible flows in astrophysics and atmosphere.
-  **Master's Student** **August 2020 - May 2021**  
**Department of Physics, Indian Institute of Technology Kanpur**  
Supervisor: Prof. Pankaj Jain, Department of Physics, IIT Kanpur  
Co-supervisor: Prof. J.S. Yadav, Department of Physics, IIT Kanpur  
Project: **Theoretical Modelling of Accretion Disk Oscillations**
  - Studied acoustic normal modes in thin accretion disks using an effective Kerr potential.
  - Derived a dispersion relation valid over a range of black hole spin parameters.

## PUBLICATIONS

### Published

1. A. Pandey, **H. Tiwari**, and K. R. Sreenivasan, Thermal convection in 1, 2, 3 and 4 dimensions, *Journal of Fluid Mechanics* (in press, 2026).
2. D. Singh, **H. Tiwari**, L. Sharma, and M. K. Verma, Scale-by-Scale Energy Transfers and Fluxes in Compressible Turbulence, *Europhysics Letters*, **154** 53001 (2026).
3. **H. Tiwari**, L. Sharma, and M. K. Verma, On the absence of the ultimate regime in turbulent thermal convection, *The Proceedings of the National Academy of Sciences*, **122** (44) e2513474122 (2025).
4. **H. Tiwari**, L. Sharma, and M. K. Verma, Compressible turbulent convection at very high Rayleigh numbers, *International Journal of Heat and Mass Transfer*, **242**, 126821 (2025).
5. D. Singh, **H. Tiwari**, L. Sharma, and M. K. Verma, Mathematical formulation of mode-to-mode energy transfers and energy fluxes in compressible turbulence, *Physical Review Fluids*, **10**, 114609 (2025).
6. L. Sharma, M. Pathak, **H. Tiwari**, and M. K. Verma, Effect of Prandtl number on turbulent compressible convection, *Physical Review Fluids*, **10**, 114611 (2025).
7. L. Sharma, M. Pathak, **H. Tiwari**, and M. K. Verma, Variation of convective heat flux imbalance with Prandtl number, *Center for Turbulence Research Annual Reports Briefs 2025*.

### Under Review

1. **H. Tiwari**, D. Singh, M. K. Verma, and R. Ranjan, Scaling in supersonic turbulence: Energy spectra and fluxes using high-fidelity direct numerical simulations, *Physical Review E*.
2. M. Pathak, L. Sharma, **H. Tiwari**, S. Aseeri, D. Keyes, and M. K. Verma, Prandtl-number dependence of positive and negative heat fluxes in turbulent compressible convection, *Phys. Rev. Fluids*.

### Under Preparation

1. **H. Tiwari**, N. Kumar, M. K. Verma, and S. Ravichandran, Direct numerical simulations of moist turbulent compressible convection.
2. **H. Tiwari**, A. Pandey, K. R. Sreenivasan, Scaling laws in four-dimensional turbulent convection.
3. D. Singh, **H. Tiwari**, and M. K. Verma, Thermalization in compressible Euler and quantum turbulence.
4. N. Kumar, **H. Tiwari**, M. K. Verma, and S. Ravichandran, Intermittency in conditionally unstable moist convection.

## CONFERENCES AND WORKSHOPS

- Workshop on “Instabilities and transitions in geophysical flows” May 2026  
**Mathematical Developments in Geophysical Fluid Dynamics trimester program**  
*Institut Henri Poincaré, Paris, France*
- “High-resolution simulations of dry and moist turbulent convection” (Oral) May 2026  
**Moist Convective Dynamics of Monsoon II**  
*International Centre for Theoretical Sciences, Bengaluru, India*
- **Indo-Japanese Workshop for CFD-based AI and HPC** December 2025  
*Indian Institute of Technology Kanpur, India*
- “Compressible turbulent convection at very high Rayleigh numbers” (Oral) November 2025  
**APS Division of Fluid Dynamics Annual Meeting**  
*Houston, Texas, USA*
- “Compressible turbulent convection at extreme Rayleigh numbers” (Poster) October 2025  
**The Variable Sun: Past, Present, and Future Perspectives**  
*Thiruvananthapuram, India*
- “Classical 1/3 Nusselt Scaling in Compressible Convection” (Oral) September 2024  
**1st European Fluid Dynamics Conference (EFDC1)**  
*Aachen, Germany*
- “Classical 1/3 Nusselt Scaling in Compressible Convection at Extreme Ra” (Oral) June 2024  
**Theoretical and Practical Perspectives in Geophysical Fluid Dynamics**  
*International Centre for Theoretical Sciences, Bengaluru, India*
- “Classical Nusselt 1/3 scaling up to  $Ra = 10^{16}$  in turbulent compressible convection” (Poster) March 2024  
**HPC Symposium**  
*Indian Institute of Technology Kanpur, India*
- “Compressible turbulent convection” (Oral) February 2024  
**Research Scholar Day**  
*Department of Physics, Indian Institute of Technology Kanpur, India*

- Scaling the Dhara solver on Frontier up to 8192 GPUs  
**Frontier Hackathon**  
*Oak Ridge National Laboratory, USA* March 2024
- **Field Theory and Turbulence**  
*International Centre for Theoretical Sciences, Bengaluru, India* December 2023
- **Turbulence: Problems at the interface of Mathematics and Physics**  
*International Centre for Theoretical Sciences, Bengaluru, India* October 2023
- Porting Gross–Pitaevskii solver to 64 Nvidia A100 GPUs  
**NSM GPU Hackathon**  
*India* November 2022
- **Summer School on Gravitational-Wave Astronomy**  
*International Centre for Theoretical Sciences, Bengaluru, India* June 2021

## TEACHING EXPERIENCE

I have assisted in the following courses at IIT Kanpur:

- PHY111A: Undergraduate Lab Aug 2025 – Nov 2025
- PHY461A/462A: Experimental Physics I/II Jan 2024 – May 2025
- NPTEL: Tapestry of Field Theory: Classical Quantum, Equilibrium, Nonequilibrium Perspectives Jan 2024 – Jan 2025
- NPTEL: Scientific Computing using Python Jun 2023 – Nov 2023
- PHY113A: Classical Electrodynamics Mar 2023 – Nov 2023
- PHY473A: Computational Physics Aug 2022 – Nov 2022
- PHY441A: Electronics Aug 2021 – May 2022

## COMPUTATIONAL SKILLS

- **Advanced:** Python, parallel programming, Numba, Paraview
- **Intermediate:** Matlab, Mathematica, CUDA
- **Basic:** C++, Julia, Fortran, R

## GRANTS AND AWARDS

- Recipient of the **NYU Postdoctoral Research and Professional Development Support Grant** (~\$2,500), Office of Postdoctoral Affairs, New York University, supporting presentation at the Supercomputing Conference 2026 (SC26).
- Recipient of the **Outstanding PhD Thesis Award** from IIT Kanpur, 2026.
- Recipient of the **2025 Division of Fluid Dynamics Enabling Award**, American Physical Society, supporting attendance at the 2025 DFD Annual Meeting, Nov 2025.
- Session chair for “Convection and Buoyancy-Driven Flows: Heat Transfer, Instabilities & Turbulence” at **APS Division of Fluid Dynamics Annual Meeting 2025** in *Houston, Texas, USA*.
- Secured an **All India Rank 74** among 15,000 applicants in the **IIT Joint Admission Test (JAM) 2019**, for admission to the M.Sc. program at the Indian Institute of Technology Kanpur.
- Recipient of **Merit cum Means Scholarship** at Indian Institute of Technology Kanpur, Aug 2019 - May 2021.
- Secured **All India Rank 307** in **Joint Entrance Screening Test (JEST) 2019**.
- Qualified **National Defence Academy (NDA)** entrance exam in 2016.

## OTHER EXPERIENCES

1. **Coordinator, Adventure Sports Club, IIT Kanpur (2024–25)**
  - Led club activities, organizing fitness programs, treks, runs, and yoga, boosting participation and visibility.
2. **Secretary, Adventure Sports Club, IIT Kanpur (2022–23, 2023–24)**
  - Assisted in organising events and managing logistics for outdoor activities and training sessions.
3. **Election Officer, Hall 7, HEC Elections 2022**
  - Managed smooth conduct of hostel elections with a voter turnout of ~85%.

## REFERENCES

1. **Prof. Katepalli R. Sreenivasan**  
Professor of Physics, Mathematical Sciences, and Engineering, New York University, USA.  
✉ [katepalli.sreenivasan@nyu.edu](mailto:katepalli.sreenivasan@nyu.edu)
2. **Prof. Mahendra K. Verma**  
Department of Physics, IIT Kanpur, India.  
✉ [mkv@iitk.ac.in](mailto:mkv@iitk.ac.in)

3. **Prof. Rajesh Ranjan**

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